

Lab task :1

DAA(LAB)

GRAPHS

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que -1 adjacency matrix and list

result:- code completely executed and got the output

```
int V, E;
int adjMatrix[MAX][MAX] = {0};
int u, v;
scanf("%d %d", &V, &E);

int adjList[MAX][MAX] = {0};
int adjListSize[MAX] = {0};

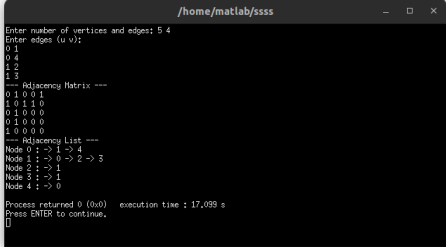
printf("Enter edges (u v):\n");
for (int i = 0; i < E; i++) {
    scanf("%d %d", &u, &v);

    adjMatrix[u][v] = 1;
    adjMatrix[v][u] = 1;

    adjList[u][adjListSize[u]++] = v;
    adjList[v][adjListSize[v]++] = u;
}

printf("Adjacency Matrix\n");
for (int i = 0; i < V; i++) {
    for (int j = 0; j < V; j++) {
        printf("%d ", adjMatrix[i][j]);
    }
    printf("\n");
}

printf("Adjacency List\n");
for (int i = 0; i < V; i++) {
    printf("Node %d : ", i);
    for (int j = 0; j < adjListSize[i]; j++) {
        printf(" -> %d", adjList[i][j]);
    }
    printf("\n");
}
```



```
Enter number of vertices and edges: 5 4
Enter edges (u v):
0 1
0 4
1 2
1 3
--- Adjacency Matrix ---
0 1 0 0 1
1 0 1 0
0 1 0 0 0
0 1 0 0 0
1 0 0 0 0
--- Adjacency List ---
Node 0 : -> 1 -> 4
Node 1 : -> 0 -> 2 -> 3
Node 2 : -> 1
Node 3 : -> 2
Node 4 : -> 3
Process returned 0 (0x0)   execution time : 17,099 s
Press ENTER to continue.
```

que-2 cyclic graph

result:- code completely executed and got the output

part 1(system safe)

```

#include <stdio.h>
#include <stdlib.h>

#define MAX 100

int dfs(int node, int visited[], int recStack[], int adjMatrix[MAX][MAX], int V) {
    visited[node] = 1;
    recStack[node] = 1;

    for (int i = 0; i < V; i++) {
        if (adjMatrix[node][i] == 1) {
            if (!visited[i] && dfs(i, visited, recStack, adjMatrix, V)) {
                return 1;
            } else if (recStack[i]) {
                return 1;
            }
        }
    }

    recStack[node] = 0;
    return 0;
}

int isCyclic(int adjMatrix[MAX][MAX], int V) {
    int visited[MAX] = {0};
    int recStack[MAX] = {0};

    for (int i = 0; i < V; i++) {
        if (!visited[i]) {
            if (dfs(i, visited, recStack, adjMatrix, V)) {
                return 1;
            }
        }
    }

    return 0;
}

int main() {
    int V, E;
    int adjMatrix[MAX][MAX] = {0};

```

```

/home/matlab/kkkk
4 3
0 1
1 2
2 3
System Safe
Process returned 0 (0x0)   execution time : 12.467 s
Press ENTER to continue.

```

part 2(deadlock detected)

```

#include <stdio.h>
#include <stdlib.h>

#define MAX 100

int dfs(int node, int visited[], int recStack[], int adjMatrix[MAX][MAX], int V) {
    visited[node] = 1;
    recStack[node] = 1;

    for (int i = 0; i < V; i++) {
        if (adjMatrix[node][i] == 1) {
            if (!visited[i] && dfs(i, visited, recStack, adjMatrix, V)) {
                return 1;
            } else if (recStack[i]) {
                return 1;
            }
        }
    }

    recStack[node] = 0;
    return 0;
}

int isCyclic(int adjMatrix[MAX][MAX], int V) {
    int visited[MAX] = {0};
    int recStack[MAX] = {0};

    for (int i = 0; i < V; i++) {
        if (!visited[i]) {
            if (dfs(i, visited, recStack, adjMatrix, V)) {
                return 1;
            }
        }
    }

    return 0;
}

int main() {
    int V, E;
    int adjMatrix[MAX][MAX] = {0};

```

```

/home/matlab/kkkk
4 3
0 1
1 2
2 0
Deadlock Detected
Process returned 0 (0x0)   execution time : 22.621 s
Press ENTER to continue.
[[[

```